The ordered Beta-distribution

Parametrisation

Let us first summarize the Beta-distribution. It has the following density

$$p(y) = \frac{1}{B(a,b)} y^{a-1}(1-y)^{b-1}, \qquad 0 < y < 1, \quad a > 0, \quad b > 0$$

where $B(a,b)$ is the Beta-function

$$B(a,b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$$

and $\Gamma(x)$ is the Gamma-function. The (re-)parameterisation used is

$$\mu = \frac{a}{a+b}, \qquad 0 < \mu < 1$$

and

$$\phi = a + b, \qquad \phi > 0,$$

as it makes

$$\mathrm{E}(y) = \mu \quad \text{and} \quad \mathrm{Var}(y) = \frac{\mu(1-\mu)}{1+\phi}.$$

The parameter ϕ is know as the *precision parameter*, since for fixed μ, the larger ϕ the smaller the variance of y. The parameters $\{a, b\}$ are given as $\{\mu, \phi\}$ as follows,

$$a = \mu\phi \quad \text{and} \quad b = -\mu\phi + \phi.$$

The **ordered Beta**-likelihood (reference here[1]) is a modification which include a derived model for $y = 0$ and $y = 1$. Let $\mu = g(\eta)$, where η is the linear predictor (see below) and $g()$ is the inverse link-function, then

$$p_o(y) = \begin{cases} 1 - g(\eta - k_1) & \text{for } y = 0 \\ (g(\eta - k_1) - g(\eta - k_2))p(y) & \text{for } 0 < y < 1 \\ g(\eta - k_2) & \text{for } y = 1 \end{cases}$$

with two offset-parameters (k_1, k_2) where $k_1 < k_2$.

Link-function

The linear predictor η is linked to the mean μ using a default logit-link

$$\mu = \frac{1}{1 + \exp(-\eta)}.$$

Hyperparameter

The hyperparameter are the precision parameter ϕ, which is represented as

$$\phi = s_i \exp(\theta_1)$$

where $s = (s_i) > 0$ is a fixed scaling. The offset parameters are expressed as

$$k_1 = \theta_1 - \exp(\theta_2), \qquad k_2 = \theta_1 + \exp(\theta_2)$$

The prior is defined on $(\theta_1, \theta_2, \theta_3)$.

[1] `https://doi.org/10.1017/pan.2022.20`

Specification

- `family="obeta"`
- Required argument: y
- Optional argument: s (argument `scale`, default all 1, $s > 0$)

Hyperparameter spesification and default values

doc `The ordered Beta likelihood`

hyper

theta1

hyperid `61101`
name `precision parameter`
short.name `phi`
output.name `precision-parameter for the obeta observations`
output.name.intern `intern precision-parameter for the obeta observations`
initial `2.30258509299405`
fixed `FALSE`
prior `loggamma`
param `1 0.1`
to.theta `function(x) log(x)`
from.theta `function(x) exp(x)`

theta2

hyperid `61102`
name `offset location`
short.name `loc`
output.name `offset location-parameter for the obeta observations`
output.name.intern `intern offset location-parameter for the obeta observations`
initial `0`
fixed `FALSE`
prior `normal`
param `0 10`
to.theta `function(x) x`
from.theta `function(x) x`

theta3

hyperid `61103`
name `offset width`
short.name `width`
output.name `offset width-parameter for the obeta observations`
output.name.intern `intern offset width-parameter for the obeta observations`
initial `0`
fixed `FALSE`
prior `loggamma`
param `1 1`

to.theta `function(x) log(x)`
from.theta `function(x) exp(x)`

survival `FALSE`

discrete `FALSE`

link `default logit loga cauchit probit cloglog ccloglog loglog`

pdf `obeta`

Example

In the following example we estimate the parameters in a simulated example.

```
library(INLA)

robeta <- function(eta, k1, k2, precision)
{
    stopifnot(k1 < k2)
    g <- function(...) inla.link.invlogit(...)
    p <- cbind(1 - g(eta - k1), g(eta - k1) - g(eta - k2), g(eta - k2))
    eta <- as.vector(eta)
    n <- length(eta)
    group <- numeric(n)
    x <- numeric(n)
    for(i in 1:n) {
        group[i] <- sample(1:3, 1, prob = p[i, ])
        if (group[i] == 1) {
            x[i] <- 0
        } else if (group[i] == 3) {
            x[i] <- 1
        } else {
            mu <- g(eta[i])
            a <- mu * precision
            b <- -mu * precision + precision
            x[i] <- rbeta(1, a, b)
        }
    }
    return (x)
}

n <- 10^4
x <- rnorm(n, sd = 0.3)
eta <- 1 + x
k1 <- -2
k2 <- 2
prec <- 2
y <- robeta(eta, k1, k2, prec)

r <- inla(y ~ 1 + x,
          data = data.frame(y, x),
          family = "obeta",
```

```
          control.inla = list(cmin = 0),
          verbose = TRUE)
summary(r)

## produce posteriors for k1 and k2
xx <- inla.hyperpar.sample(10^5, r, intern = FALSE)
loc <- xx[, "offset location-parameter for the obeta observations"]
width <- xx[, "offset width-parameter for the obeta observations"]
k1.mc <-  loc - lwidth
k2.mc <- loc + lwidth
par(mfrow = c(1, 2))
hist(k1.mc, prob=TRUE, n = 100); abline(v=k1, lwd = 5, col = "blue")
hist(k2.mc, prob=TRUE, n = 100); abline(v=k2, lwd = 5, col = "blue")
```

Notes

None.